Lecture-14
Conservation Laws for Control Volumes II: Conservation of Mass

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Learning Objectives

By the end of this lecture, you should be able to

1. State the integral form of the **control volume conservation of mass equation** from first principles
2. Do some applications of the conservation of mass equation for **some engineering problems**

Conservation of mass

- Barring the addition of a substance from some external source, the **mass of the system** remains constant.

$$\left(\frac{dM}{dt}\right)_{\text{system}} = \frac{\partial}{\partial t}\int_{\text{CV}} \rho \, d\forall + \int_{\text{CS}} \rho \vec{V} \cdot d\vec{A}$$

- **Rate of increase of mass in the control volume is balanced by the net flux of mass through the control surface**.

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Revisiting the learning objectives

1. State the integral form of the **control volume conservation of mass equation** from first principles
2. Do some applications of the conservation of mass equation for **some engineering problems**